

Subject Name: Software Engineering Laboratory

Subject Code: CS 314

B.Tech (CSE) Fifth Semester (Core)

Important Notes for Lab Copy Submission

1. Lab Copy Must Be Handwritten

- All explanations should be written neatly by hand except diagrams. Typed or printed content will not be accepted.
- All UML-related experiments and diagrams (Use Case Diagram, Class Diagram, ER Diagram, etc.) will be performed using **StarUML** software.

2. Attendance Policy

- Each lab session is 3 hours long and will be counted as 1 attendance hour.
- A total of 3 attendance hours will be recorded per full lab session.

3. Continuous Evaluation Criteria

The performance of each student in the lab will be continuously evaluated based on the following components:

- Lab Copy
 - Neatness, completeness, correctness and accuracy of diagrams and write-ups.
 - Proper maintenance of the index and adherence to lab guidelines.
- Execution of Experiments
 - Practical understanding and successful implementation of diagrams using StarUML/Other as instructed
 - Active participation during lab sessions and timely completion of assigned tasks.
- Monitoring of attendance as mentioned in Sl. No.2

4. Proper Indexing

- An index must be properly maintained at the beginning of the lab copy.
- Each entry should include:
 - Serial Number
 - Experiment Name
 - Date experiment& submission
 - Page Number, Remarks etc.

5. Cover Page & First Page Requirements

The Cover Page and First Page (before the Index) must contain the following details:

- Subject Name
- Subject Code
- Name of the Student

- Scholar ID
- Semester
- Section
- Department Name
- Institute Name
- Session

Problem statement-1: Hospital Management System (HMS)

Overview

The **Hospital Management System (HMS)** is a powerful, flexible, and user-friendly web-based application designed to streamline hospital operations and enhance overall efficiency in multi-specialty hospital environments. It aims to support various administrative, clinical, and financial tasks through a centralized platform, enabling a smooth flow of information across departments and users.

HMS facilitates efficient hospital administration by automating processes such as patient registration, doctor allocation, staff management, billing, report generation, and record access. It also supports decision-making by providing real-time access to critical data for hospital staff and management.

Key Features and Functional Requirements

- **Data Entry and Management:**
 - Secure data entry and management for patients, doctors, staff, and administrators.
 - Authorized personnel can access and update data as per their roles.
- **Patient Record Access:**
 - Patients can log in to view and download their own medical reports and bills.
 - Doctors can view and manage only their assigned patients.
- **Billing and Payments:**
 - The system supports billing management with real-time updates after each transaction.
 - Automatically generates receipts and stores billing history per patient.

Inputs

- **Hospital Information:** Name, address, branch details, departments, facilities.
- **Employee Information:** Name, employee ID, address, contact details.
- **Doctor Information:** Name, employee ID, specialization, degrees, contact details.
- **Patient Information:** Name, patient ID, age, address, contact details, payment details, medical records.

Outputs

- Patient details and medical history
- List of patients under a specific doctor
- Bill/receipt generation
- Medical report generation and download

System Constraints

- All users must register through the web interface.
- Each user must log in using a **unique user ID and password**.
- Patients are permitted to access **only their own information**.
- Doctors can only view the list of patients assigned to them.
- Patients must be able to **download all medical reports and billing information** through their accounts.
- Any addition, deletion, or update in the system must be **reflected immediately** in the database.

Security and Access Control

- **Role-based access control:** Different levels of access for administrators, doctors, staff, and patients.
- **Data privacy:** Patients must not be able to view or access other patients' information.

This system will significantly reduce paperwork, improve the accuracy and accessibility of records, and enhance the overall patient experience by offering seamless digital services.

Objectives:

The primary objectives of this project are:

1. To draw a Use Case Diagram

- a. To visually represent the interactions between different types of users (actors) such as patients, doctors, hospital staff, and administrators with the system.
- b. To identify and define the functional requirements of the Hospital Management System through actor-use case relationships.

2. To draw a Class Diagram

- a. To model the static structure of the system by showing the system's classes, attributes, operations, and the relationships among the classes.
- b. To represent how the data and behavior of the hospital entities (e.g., Patient, Doctor, Appointment, Bill) are organized.

3. To draw an Entity-Relationship (ER) Diagram

- a. To represent the logical structure of the database by identifying entities, attributes, and relationships.
- b. To assist in the design of the relational database that supports the backend of the Hospital Management System.

4. (We will add more)